

PROJECT 4 (BUMPER RALLY)

CHAUGHN ROBIN



How To Play

Main Menu Scene –

**On launch, the game begins on the main menu screen.
The options are ‘Host, ‘Join, and ‘Exit’.**

- Press **Host** to host a game session. This must be done before other players can join.
- **Press Join to join a currently hosted session.**
- **Press Start Game to start the currently hosted session.**
- **Press Exit to exit the game.**

Controls –

- **‘WASD keys or ↑←↓→ keys’ are used for the movement controls.**
- **W / ↑** is used to drive forwards.
- **S / ↓** is used to reverse,
- **D / →** rotates the player to the right.
- **A / ←** rotates the player to the left.
- **Hold X** to start your bumper car’s engine. On successful start, you will hear the engine idle and see the headlights turn on.
- **SHIFT** gives you a turbo boost and switches your camera while you speed up for close up views of the action!
- **SPACE** applies the brakes and allows for slick drifting.
- **ESC** opens or closes the menu overlay.
- The controls can also be found by using the menu overlay’s **Controls** tab.

Game Multiplayer Scene –

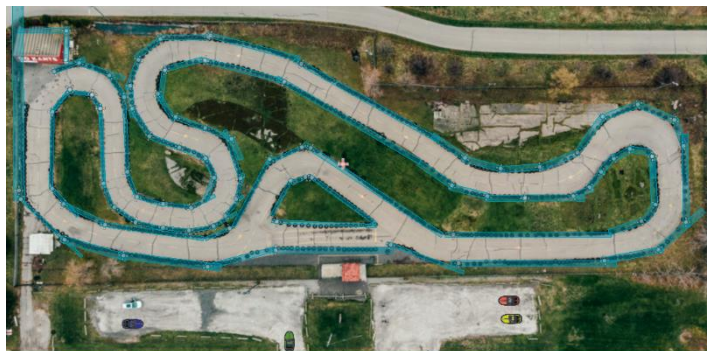
On start, you may be wondering where you are. In the top left corner of the screen, you will see text reading ‘Spawnpoint’. All go-karts spawn in the red ‘GO KARTS’ building in the top left. Hold X to start your engine, then drive out onto the track!

The last car standing wins!

Obstacles –

Bumper Cars are one danger to be on the watch for. Colliding with enemy cars will cause both damage. Look out!

Boundaries also cause the player to take damage. Try to avoid bumping into them.



When a **Bumper Car**'s health reaches 0, they are no longer operational.

Players –

Players drive bumper cars with many different features. Currently, the game handles 2-4 players.



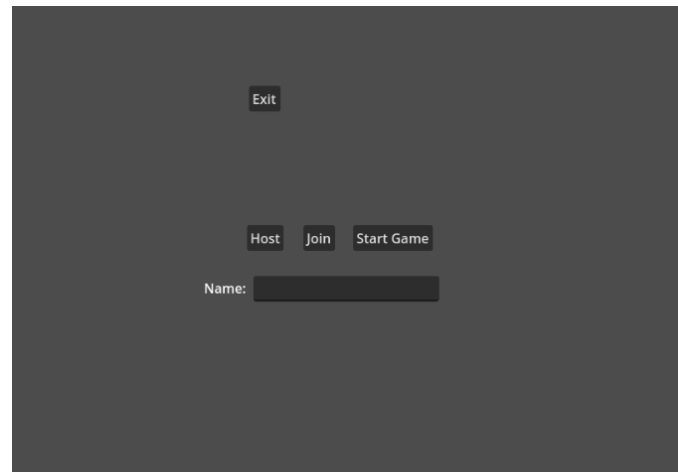
Each player operates a different color bumper car. When turned on, the bumper car's headlight flick on! Each bumper car has a range of sounds from idling, revving, to different crash sounds. They also have their own camera for high-paced exciting angles.

Cars have some cool details like when they brake on the move, they leave skid marks behind on the road. Listen in and you can hear that asphalt burning up rubber!



Client and Server –

The client and server work using one MultiplayerScene. Based on what button is pressed, that decides what result you get. Host creates a host server and join creates a peer client to join the server. Start game sends the start RPC to all, changing the server and clients to the next scene, the game scene.



Results of the Playtesting

Rev2:

- Had many issues synchronizing player states across server
- Initially, I was changing texture and trying to sync that by RPC, player was visible to everyone but the correct client. Changed the functionality of turning on the headlight to be a Sprite2D that is hidden, and then we just share its visible state (Boolean), much easier than trying to update texture by RPC.
- With more time, would have done more RPC calls for proper synchronization of all details, like skid marks.

Rev1:

- Initial issues with getting multiplayer set up working
- Several issues with audio looping infinitely (a car crashes into a wall, what will make it stop playing that crash audio if it plays based on contact with collision?)
- Several bumper car speed & resistance changes
- Also, maps are extremely easy to produce. All you need is a top-down picture and create the boundaries for it. This will allow for lots of modularity.